

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) The following question is based on information from Kondo Y. and Kashiwagi A., Experimentally induced autotetraploidy and allotetraploidy in two Japanese pond frogs, *Journal of Herpetology* 38(3):381–92 (2004). Two researchers experimentally formed tetraploid frogs by fertilizing diploid eggs from *Rana porosa brevipoda* with diploid sperm from *Rana nigromaculata*. When they allowed the tetraploid frogs to mate with each other, most of the offspring that survived to maturity were tetraploid, with chromosome sets of both diploid parent species. Based on these results, if this type of tetraploid formed in the wild, what would be the most likely result?
 - A) The tetraploids would be reproductively isolated from both parent species.
 - B) The two parent species would interbreed and fuse into one species.
 - C) The two parent species would recognize each other as mates.
 - D) The tetraploids would invade the habitats of both parental species and cause them to go extinct.
- 2) New species can sometimes arise via hybridization and subsequent polyploidy. Which of the following describes the likely geographic and reproductive relationships between the hybridizing species?
 - A) The two species are sympatric, but breed during different times of the year.
 - B) The two species are allopatric and breed at different times of the year.
 - C) The two species can be either allopatric or sympatric, but breed during the same seasons of the year.
 - D) The two species are sympatric and breed during the same seasons of the year.
- 3) According to the concept of punctuated equilibrium, the "sudden" appearance of a new species in the fossil record means that _____.
 - A) the species will consequently have a relatively short existence compared with other species
 - B) speciation occurred rapidly in geologic time
 - C) the species is now extinct
 - D) speciation occurred in one generation
- 4) Which of the following factors is the primary criterion for determining species boundaries when applying the biological species concept?
 - A) geographic isolation
 - B) gene flow
 - C) morphological similarity
 - D) niche differences
- 5) Dog breeders maintain the purity of breeds by keeping dogs of different breeds apart when they are fertile. This kind of isolation is most similar to which of the following reproductive isolating mechanisms?
 - A) temporal isolation
 - B) habitat isolation
 - C) gametic isolation
 - D) behavioral isolation

- 6) Almost all peppered moths in England were light colored before the Industrial Revolution. However, some dark forms were collected—the dark color is determined by a single, dominant allele of one gene. The bark of trees became darker during the Industrial Revolution in areas where heavy coal use in factories caused pollution. By about 1900, approximately 90% of the moths around industrial areas were dark, whereas light-colored moths were still abundant elsewhere. Apparently, birds could readily find the light moths against the dark trees in industrial areas and, therefore, were eating more light moths. Recently, use of cleaner fuels has resulted in tree bark becoming lighter again and the dark-colored moths have been disappearing. Which of the following statements best explains whether we should consider the forms different species?
- A) They are different species—as shown by the fact that natural selection has affected the frequency of the two forms.
 - B) They are NOT different species—as shown by the fact that they live in the same habitats.
 - C) They are different species—as shown by the fact that the two forms are reproductively isolated by habitat.
 - D) They are NOT different species—as shown by the fact that they can interbreed.
- 7) Which of the following characteristics is a postzygotic barrier that prevents hybrids between two species from producing offspring?
- A) hybrid breakdown
 - B) increased hybrid vulnerability
 - C) hybrid gamete isolation
 - D) increased hybrid fertility
- 8) Beetle pollinators of a particular plant are attracted to its flowers' bright orange color. The beetles not only pollinate the flowers, but they mate while inside the flowers. A mutant version of the plant with red flowers becomes more common with the passage of time. A particular variant of the beetle prefers the red flowers to the orange flowers. Over time, these two beetle variants diverge from each other to such an extent that interbreeding is no longer possible. Which of the following possible causes of speciation best describes the beetle example and what factor has driven it?
- A) sympatric speciation; allopolyploidy
 - B) allopatric speciation; behavioral isolation
 - C) allopatric speciation; ecological isolation
 - D) sympatric speciation; habitat differentiation
- 9) A female fly, full of fertilized eggs, is swept by high winds to an island far from the mainland. She is the first, and only, fly to arrive on this island. Thousands of years later, her numerous offspring occupy the island, but none of them resembles her. There are, instead, several species and their geographic ranges overlap. Each species eats a different kind of food. None of the species can fly and the balancing organs of males differ among species and are now used in courtship displays. Fly species W produces fertile offspring with species Y, but not with species X or Z. Furthermore, X and Z do not hybridize. Which of the following statements about species W and Y is most likely to be accurate?
- A) Species W and X have more genetic similarity with each other than either does with the other two species.
 - B) Species W and Y will both be selected to avoid mating with each other.
 - C) Species W and Y may fuse into a single species if their hybrids remain fertile over the course of many generations.
 - D) Species W and Y have genomes that are too dissimilar for successful meiosis to occur in hybrid flies.
- 10) House finches (birds) occurred only in western North America until 1939, when a few individuals were released in New York City. These individuals established a breeding population and gradually expanded their range. The western population also expanded its range somewhat eastward, and the two populations have recently come in contact. Which of the following terms best describes the process that had occurred if the two forms did not interbreed when their expanding ranges met?
- A) allopatric speciation
 - B) prezygotic isolation
 - C) reinforcement
 - D) sympatric speciation

- 11) Which of the following organisms would be LEAST able to cross a geographic barrier of water that could lead to allopatric speciation?
- A) birds
 - B) plants with wind-dispersed pollen
 - C) mountain lions
 - D) small rodents
- 12) A team of scientists is trying to understand the evolutionary history of three species of grasses. They hypothesize that Species T is a descendent of a hybrid between Species R and Species S. Which of the following experimental approaches would best help elucidate the evolutionary history of these three species?
- A) Hybridize Species R and S; induce tetraploidy in the hybrids; and compare the hybrids to Species T.
 - B) Hybridize Species R and S; plant the hybrid offspring in environmental conditions similar to those of Species R; follow the morphological and genetic changes in the offspring of the hybrids; and compare any observed changes to Species R.
 - C) Sequence the DNA of all three species and determine which alleles are unique to each species.
 - D) Hybridize Species R and S; plant the hybrid offspring in environmental conditions resembling those of Species R, S, and T; follow morphological and genetic changes in the offspring of the hybrids; and compare the observed changes to Species T.
- 13) Lions and tigers sometimes breed in zoos, but hybrids have not been seen in nature and even in zoos they do not reproduce well. Which of the following statements is the best conclusion that can be derived from this information?
- A) The morphological or ecological species concepts must be used rather than the biological species concept because there could be gene flow between the species in nature.
 - B) The biological species concept can be applied because species are, in part, defined by the inability to produce viable, fertile offspring with other species.
 - C) The biological species concept is invalid because hybrids can be formed.
 - D) Lions and tigers are not actually different species because they can produce viable offspring in captivity.
- 14) Two species of frogs belonging to the same genus occasionally mate, but the embryos stop developing after a day and then die. Which of the following mechanisms keep the species separate?
- A) gametic isolation
 - B) reduced hybrid viability
 - C) reduced hybrid fertility
 - D) hybrid breakdown
- 15) Which of the following outcomes is most likely in hybrid zones where reinforcement of isolating mechanisms is occurring?
- A) increased gene flow between distinct gene pools
 - B) a decrease in the genetic distinctness of two gene pools
 - C) speciation
 - D) formation of a stable hybrid zone
- 16) Males of different species of the fruit fly *Drosophila* that live in the same parts of the Hawaiian Islands have different elaborate courtship rituals. These rituals involve fighting other males and making stylized movements that attract females. What type of reproductive isolation does this represent?
- A) gametic isolation
 - B) habitat isolation
 - C) behavioral isolation
 - D) temporal isolation
- 17) Plant species A has a diploid chromosome number of 12. Plant species B has a diploid number of 16. A new species, C, arises as an allopolyploid from A and B. The diploid number for species C would probably be
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- A) 56
 - B) 28
 - C) 14
 - D) 16

18) Which of the following factors would *not* contribute to allopatric speciation?

- A) The separated population is small, and genetic drift occurs.
- B) Gene flow between the two populations is extensive.
- C) The isolated population is exposed to different selection pressures than the ancestral population.
- D) Different mutations begin to distinguish the gene pools of the separated populations.

19) Bird guides once listed the myrtle warbler and Audubon's warbler as distinct species. Recently, these birds have been classified as eastern and western forms of a single species, the yellow-rumped warbler. Which of the following pieces of evidence, if true, would be cause for this reclassification?

- A) The two forms have many genes in common.
- B) The two forms live in similar habitats and have similar food requirements.
- C) The two forms interbreed often in nature, and their offspring survive and reproduce well.
- D) The two forms are very similar in appearance.

20) The *largest* unit within which gene flow can readily occur is a _____.

- A) species
- B) hybrid
- C) genus
- D) population

Answer Key

Testname: UNTITLED1

- 1) A
ID: bio12-- 24.1-28
- 2) D
ID: bio12-- 24.1-33
- 3) B
ID: bio12-- 24.1-54
- 4) B
ID: bio12-- 24.1-19
- 5) B
ID: bio12-- 24.1-13
- 6) D
ID: bio12-- 24.1-6
- 7) A
ID: bio12-- 24.1-17
- 8) D
ID: bio12-- 24.1-34
- 9) C
ID: bio12-- 24.1-53
- 10) A
ID: bio12-- 24.1-26
- 11) D
ID: bio12-- 24.1-24
- 12) D
ID: bio12-- 24.1-59
- 13) B
ID: bio12-- 24.1-11
- 14) B
ID: bio12-- 24.1-15
- 15) C
ID: bio12-- 24.1-47
- 16) C
ID: bio12-- 24.2-2
- 17) B
ID: bio12-- 24.2-6
- 18) B
ID: bio12-- 24.2-5
- 19) C
ID: bio12-- 24.2-4
- 20) A
ID: bio12-- 24.2-1